

Nihal Kumar Singh

Full Stack Developer — Hisua, Bihar – 805103

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Portfolio

PROFESSIONAL SUMMARY

Full Stack Developer (React.js, Node.js, TypeScript, Django) who has independently designed, built, and deployed many production-style applications spanning full-stack web platforms, Kubernetes-based cloud deployments, and CI/CD automation pipelines. Delivered systems covering database design, multi-layer authentication, 68+ automated tests, and infrastructure-as-code (Terraform) on AWS, Docker, and Kubernetes. Strong foundation in DSA and OOP with proven ability to own a project end-to-end from database schema to deployment and monitoring.

TECHNICAL SKILLS

- **Languages:** Java, JavaScript (ES6+), TypeScript, SQL
- **Frontend:** React.js (Vite), Redux, Tailwind CSS, Framer Motion, HTML5, CSS3, Responsive Web Design
- **Backend:** Node.js, Express.js, Django, RESTful APIs, JWT Authentication, Socket.IO, Microservices
- **DevOps & Cloud:** AWS, Docker, Kubernetes (EKS), Jenkins, GitHub Actions, Terraform, CI/CD, Nagios, Prometheus, Grafana, Shell Scripting, Linux
- **Core CS:** Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS

PROJECTS

- **PropSync – Property Rental & Maintenance Management Platform** Jun 2026 – Jul 2026
TypeScript, React, Node.js, Express, MongoDB Atlas, JWT, Socket.IO, Recharts [GitHub Repo](#)
 - **MERN Full Stack App:** Architected a full-stack MERN platform (React 18, TypeScript, Node.js, Express, MongoDB Atlas) serving 4 distinct role-based dashboards (Admin, Owner, Tenant, Maintenance Staff) through a single RBAC-secured codebase.
 - **Security:** Engineered a 3-layer authentication system combining JWT, bcrypt password hashing, TOTP-based 2FA, and SendGrid email OTP verification to secure all 4 user roles.
 - **Real-Time Features:** Delivered a Socket.IO notification layer pushing live updates across 3 workflows – maintenance requests, amenity bookings, and dashboard metrics – eliminating manual page refreshes.
 - **Performance:** Streamlined app performance with React Query caching, optimistic UI updates, and request deduplication, and enabled serverless-compatible file uploads (Multer) across a Vercel + Render production deployment.
- **Intelligent Internship Recommendation System** Oct 2025 – Apr 2026
React.js, Tailwind CSS, Django, PostgreSQL, Redis, Docker, Kubernetes (EKS), Prometheus, Grafana [GitHub Repo](#)
 - **Full Stack Platform:** Designed an end-to-end internship marketplace spanning 3 distinct portals (student, recruiter, admin), integrating a React frontend with a Django REST backend and PostgreSQL database.
 - **AI Recommendation Engine:** Built a custom ML recommendation engine (Cosine Similarity + VSPS trust scoring) using Scikit-Learn, NumPy, and Pandas, ranking internship matches on a normalized 0–1 relevance scale.
 - **Recruiter & Admin Dashboards:** Developed 2 analytics-heavy dashboards covering application tracking, candidate ranking, organization verification, and 2 risk controls (2FA enforcement, auto-approval).
 - **Testing & CI/CD:** Authored a 68-test automated suite (backend API, frontend, E2E) and automated deployment through a GitHub Actions CI/CD pipeline with coverage reporting and security scanning.
 - **Cloud Infrastructure:** Provisioned a high-availability AWS EKS cluster with IAM/OIDC-integrated Kubernetes Service Accounts for secure, password-less pod access, and configured the EBS CSI driver for dynamic persistent volume provisioning.
 - **Traffic & Autoscaling:** Configured the AWS Load Balancer Controller to automate public ALB creation via Kubernetes Ingress, and implemented Horizontal Pod Autoscaling to scale backend pods when CPU exceeds 70% or memory exceeds 80%.
 - **Stateful Data Layer:** Deployed PostgreSQL and Redis as Kubernetes StatefulSets, containerized frontend/backend services as custom Docker Hub images, and ran production database migrations seeded with extensive dummy data.
 - **Observability & Deployment:** Installed a Metrics Server for cluster-wide CPU/RAM tracking, configured Prometheus to scrape time-series metrics every 15 seconds, and visualized pod stability and request rates in Grafana; automated the full build-and-deploy cycle via GitHub Actions in under 4 minutes.
- **AI Resume Intelligence System** Jan 2025 – Feb 2025
Java, OOP, Java Swing, NLP, DSA [GitHub Repo](#)
 - **Desktop ATS System:** Constructed a desktop-based ATS system in Core Java following OOP principles, cleanly separating the GUI (Java Swing) from business logic for maintainability.
 - **NLP Pipeline:** Devised a custom NLP processing pipeline covering tokenization, stop-word filtering, and semantic analysis to parse and interpret resume content.
 - **Scoring Engine:** Implemented DSA-driven algorithms for skill-frequency analysis and built a 5-factor scoring model to rank resumes by relevance.

TRAINING

- **GeeksForGeeks** May 2024 – Aug 2024
DSA Summer Internship Training — Java, JavaScript Online
 - **DSA Coverage:** Completed a 3-month, intensive curriculum spanning fundamentals to advanced data structures and algorithms, with emphasis on time/space complexity optimization.
 - **Sudoku Solver:** Solved a 9x9 Sudoku Solver challenge using JavaScript backtracking with recursion and constraint pruning; paired it with an interactive HTML/CSS UI.

CERTIFICATIONS

- **Java Programming:** GeeksForGeeks – Sep 2025
- **Cloud Computing:** NPTEL – Oct 2024
- **Data Structures and Algorithms:** GeeksForGeeks – Sep 2024

EDUCATION

- **Lovely Professional University** 2022 – 2026
B.Tech in Computer Science and Engineering Phagwara, Punjab
 - **CGPA:** 7.66
- **Jeevan Jyoti Public School** 2019 – 2021
12th CBSE Nawada, Bihar
 - **Percentage:** 73.6%
- **Jeevan Jyoti Public School** 2018 – 2019
10th CBSE Nawada, Bihar
 - **Percentage:** 91.2%